OPLOGS-01: OpenPipeline Fundamentals

> **Series:** OPLOGS — OpenPipeline Logs | **Notebook:** 1 of 8 | **Created:** December 2025 | **Last Updated:** 10/02/2026

Understanding the Unified Data Ingestion Framework

This notebook introduces OpenPipeline, Dynatrace's unified data processing framework for logs, traces, metrics, and events.

> 🆕 **New Addition (March 2026)**
>
> **Companion Series**
> - **OPMIG** — Migrating from classic logs to OpenPipeline? Start with **OPMIG-01: Why Migrate**
> - **OPIPE** — Ready to process spans, metrics, and events? Continue to **OPIPE-01: OpenPipeline as a Multi-Scope Platform**
> - **SPANS** — Need to query and analyze distributed traces? See **SPANS-01: Fundamentals**

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Prerequisites

- ✅ Access to a Dynatrace environment with log data
- ✅ DQL query permissions (viewer role minimum)
- ✅ Basic understanding of log management concepts

1. What is OpenPipeline?

OpenPipeline is Dynatrace's unified data ingestion and processing framework that replaces classic log ingestion. It provides:

- **Unified Processing**: Single framework for logs, metrics, traces, and business events
- **Real-time Transformation**: Parse, enrich, mask, and route data at ingestion
- **Grail Storage**: Direct integration with Dynatrace's data lakehouse
- **Flexible Routing**: Send data to different buckets with custom retention
- **Cost Control**: Drop unnecessary data before storage

OpenPipeline vs the Classic Log Pipeline

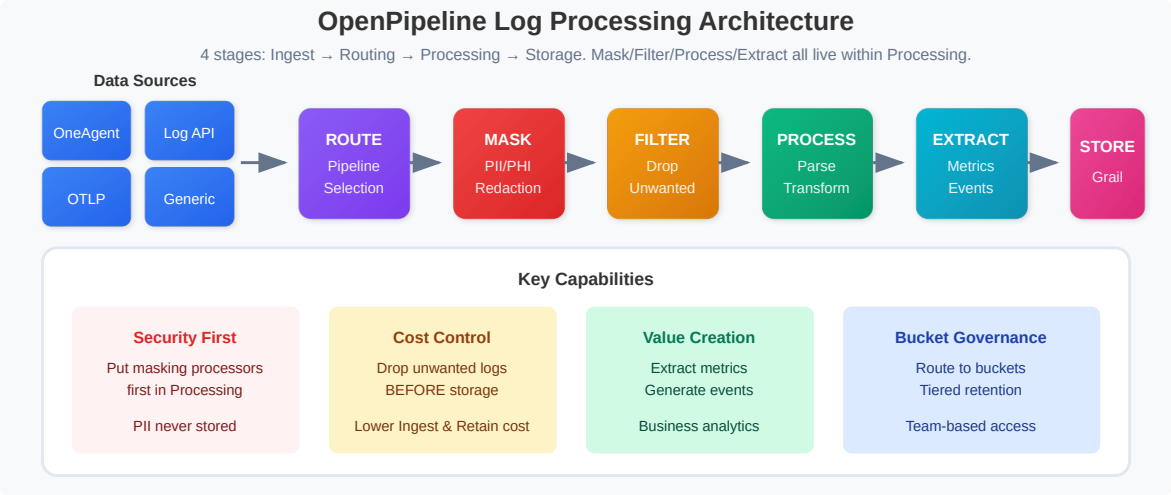
On Grail, OpenPipeline replaces the **classic log pipeline** — log processing rules under *Settings > Log Monitoring*. Both process records **before** they are stored; the differences are scope, language and structure:

Feature	Classic log pipeline	OpenPipeline
Data Processing	Processing rules at ingest, before storage	Pipelines at ingest, before storage
Signal types	Logs and business events	Logs, spans, metrics, events, business events, and more
Rule language	Processing-rule commands with DPL	DQL processors with DPL, plus no-code processors

Feature	Classic log pipeline	OpenPipeline
Routing	One ordered list of rules	Dynamic routing to many pipelines, pipeline groups
Data Masking	Mask / drop attributes in a rule	DQL processor with DPL (`replacePattern`), `replaceString` , `ipMask`
Extraction	Log metrics	Metric, Davis event, business event, SDLC event, Smartscape stages
Retention	Bucket per record	Bucket assignment stage, or No storage assignment

> **Sources:** Log processing with classic pipeline (DT docs) — *"Log processing occurs as log data arrives in the Dynatrace SaaS environment and before it is written to disk (stored)."*

2. OpenPipeline Architecture



> **Stage model (verified 09/24/2026):** data flows Ingest → Routing → pipeline → Storage. Inside a pipeline, per *Processing in OpenPipeline*, "The sequence of stages is fixed for all pipelines and cannot be modified." Masking, dropping, parsing and transformation are processors in the first stage, **Processing**; **Bucket assignment** comes seventh, and **Metric extraction**, **Davis** and **Data extraction** run after it. The diagram's Mask / Filter / Process boxes are all inside that first stage.
>
> **Sources:** Processing in OpenPipeline (DT docs)

3. Exploring Your OpenPipeline Data

Let's start by discovering what data sources and pipelines are active in your environment.

```
// Discover data sources feeding OpenPipeline
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {dt.openpipeline.source}
| sort log_count desc
```

```
// See which pipelines are processing your logs
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {dt.openpipeline.pipelines}
| sort log_count desc
```

```
// Check storage bucket distribution
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {dt.system.bucket}
| sort log_count desc
```

4. Key OpenPipeline Fields

OpenPipeline adds metadata fields to every log record:

Pipeline Metadata

Field	Description	Example
`dt.openpipeline.source`	How the log was ingested	`oneagent`, `/api/v2/logs/ingest`, `/api/v2/otlp/v1/logs`
`dt.openpipeline.pipelines`	Pipeline(s) that processed the log	`["logs:pipeline_Default_Pipeline_2798"]`
`dt.system.bucket`	Storage bucket name	`default_logs`, `custom_logs`

Core Log Fields

Field	Description
`timestamp`	When the log was generated
`content`	The log message body
`loglevel`	Log severity (ERROR, WARN, INFO, DEBUG, NONE)
`status`	Severity group derived from `loglevel`: ERROR (SEVERE, ERROR, CRITICAL, ALERT, FATAL, EMERGENCY), WARN, INFO (INFO, TRACE, DEBUG, NOTICE), NONE — use it for error-class counts
`log.source`	Source identifier (e.g., "Container Output")
`log.iostream`	Stream type (stdout, stderr)

```
// View a sample log record with all key fields
fetch logs, from: now() - 1h
| fields timestamp, content, loglevel, status, log.source, log.iostream,
         dt.openpipeline.source, dt.openpipeline.pipelines, dt.system.bucket
| limit 5
```

```
// Analyze log levels in your environment
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

5. Data Sources Explained

OneAgent (`oneagent`)

Logs collected automatically by Dynatrace OneAgent from:
- Container stdout/stderr
- Process log files
- System logs

Log Ingest API (`/api/v2/logs/ingest`)

Logs sent directly via the Dynatrace API:
- Custom application logs

- Third-party integrations
- Cloud provider logs (AWS, Azure, GCP)

Delivery reliability

The Log Ingest API acknowledges accepted data with **HTTP 204 No Content** (empty body). When a request fails on a *retryable* response code (each endpoint documents which codes are retryable), the **client is responsible for retrying with exponential backoff** — Dynatrace does not retry on the sender's behalf. The ActiveGate that serves the endpoint buffers ingested data to an on-disk queue (default **300 MB**, configured by `disk_queue_max_size_mb`; location by `disk_queue_path`) and forwards it to Dynatrace in batches; when that queue fills, the API returns `503 Usable space limit reached` as a back-pressure signal — increase `disk_queue_max_size_mb` if this recurs under normal load. Cloud-forwarder integrations (e.g. Amazon Data Firehose) implement their own buffering and smart-retry against the OpenPipeline endpoint.

OTLP (`/api/v2/otlp/v1/logs`)

OpenTelemetry Protocol logs:
- OpenTelemetry Collector
- Fluent Bit with OTLP output
- Custom OTLP exporters

```
// Compare volume by data source
fetch logs, from: now() - 24h
| summarize {
    log_count = count(),
    unique_hosts = countDistinct(dt.entity.host)
  }, by: {dt.openpipeline.source}
| sort log_count desc
```

```
// Logs per hour by source (trend analysis)
fetch logs, from: now() - 24h
| makeTimeseries {log_count = count()}, by: {dt.openpipeline.source}, interval:
1h
```

6. Pipeline Stages Overview

A record first passes **Routing**, which picks the pipeline (a dynamic DQL matcher, or a static route for custom ingest sources). Inside the pipeline it then runs through a **fixed** sequence of ten stages — the order cannot be changed:

#	Stage	What it does in a log pipeline	Processors	Executed
1	**Processing**	Parse values into fields, transform the schema, filter records, mask sensitive data	DQL, Add fields, Remove fields, Rename fields, Drop record, GeoIP lookup (Early Access), Inline lookup	All matches
2	Smartscape node	Extract Smartscape nodes from matching records	Smartscape node	All matches
3	Smartscape edge	Extract Smartscape edges from matching records	Smartscape edge	All matches
4	Permission	Apply a security context to matching records	Set security context	First match only
5	Product allocation	Assign product or application usage	DPS Cost Allocation - Product	First match only
6	Cost allocation	Assign cost-center usage	DPS Cost Allocation - Cost Center	First match only
7	**Bucket assignment**	Choose the Grail bucket — or not to store the record	Bucket assignment, No storage assignment	First match only
8	Metric extraction	Extract metrics from matching records	Counter metric, Histogram metric, Value metric	All matches
9	Davis	Extract a Davis event and re-ingest it into another pipeline	Davis event	All matches
10	Data extraction	Extract a business event or SDLC event and re-ingest it	Business event, Software development lifecycle event	All matches

Masking, dropping and parsing are not separate stages: they are processors in stage 1, and within a stage they run in the order you place them. Storage in the chosen Grail bucket happens after the pipeline, so a record's retention is set by stage 7 — which is also why the extraction stages (8–10) still see records assigned to **No storage assignment**.

> **Sources:** Processing in OpenPipeline (DT docs) — *"each processor output becomes the input for the next one"*, OpenPipeline processing examples (DT docs) — *"The record continues through all pipeline stages,*

> *including metric extraction, and is not stored only at the end."*

7. Environment Summary

Let's get a complete picture of your OpenPipeline environment:

```
// Complete environment summary
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    unique_sources = countDistinct(dt.openpipeline.source),
    unique_buckets = countDistinct(dt.system.bucket),
    unique_hosts = countDistinct(dt.entity.host),
    error_count = countIf(loglevel == "ERROR" OR status == "ERROR"),
    warn_count = countIf(loglevel == "WARN" OR status == "WARN")
  }
```

```
// Top log sources by entity
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.host)
| summarize {log_count = count()}, by: {host.name, log.source}
| sort log_count desc
| limit 15
```

📝 Summary

In this notebook, you learned:

✅ **What OpenPipeline is** - Dynatrace's unified data processing framework
✅ **Architecture** - Data flow from sources through processing to Grail
✅ **Key fields** - `dt.openpipeline.source`, `dt.openpipeline.pipelines`, `dt.system.bucket`
✅ **Data sources** - OneAgent, Log Ingest API, OTLP
✅ **Pipeline stages** - a fixed sequence: Processing (mask, drop, parse, transform) → … → Bucket assignment → Metric / Davis / Data extraction

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➡️ Next Steps

Continue to **OPLOGS-02: Migration Guide** to learn how to migrate from classic log ingestion to OpenPipeline v2.0.

📚 References

- OpenPipeline (DT docs)
- Processing in OpenPipeline (DT docs)
- Grail (DT docs)
- Dynatrace Query Language (DT docs)
- Automatic log enrichment (DT docs) — *"a status attribute is created with a value that is a sum of loglevel values based on the following grouping"*
- Log ingestion API (DT docs) — delivery reliability: HTTP 204 success, retryable codes + exponential backoff, ActiveGate disk-queue buffering, `503 Usable space limit reached`

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